Heterogeneous Willingness-to-Pay for EVs

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July 24, 2025

Motivation

- Transportation accounts for 25% of global CO_2 emissions.
- EVs offer a promising strategy to substantially reduce emissions.
- Governments have implemented various policies to promote EV adoption.
- Effective policy design requires insight into how vehicle attributes and demographic heterogeneity shape consumer preferences for EVs.
- However, empirical evidence on the structure of EV demand remains limited.

Research Question

- **Main Question:** What is U.S. consumers' willingness-to-pay (WTP) for EVs and hybrids? How does WTP vary with vehicle attributes and household demographics?
- I estimate a BLP model with micro-moments to quantify how **EV attributes** (battery capacity and driving range) and **demographic characteristics** shape preferences for EVs.

Preview of Results

- **Substantial heterogeneity** in consumer valuation of EVs.
- **Technical features** like battery capacity and driving range **increase utility**, but:
  - A **large baseline disutility** for EVs persists, especially among:
    - **Rural**
    - **Low-income**
    - **Non-college-educated** households
- **Urban** and **college-educated** consumers show **higher EV valuations**.
- **High-income households** exhibit:
  - Strong EV interest
  - Lower price sensitivity

Related Literature

- Discrete Choice Demand Estimation:
  - Berry, Levinsohn, and Pakes (1995): Random coefficient logit (BLP) model for differentiated product markets.
  - Petrin (2002): Use micro-moments in heterogeneous coefficients demand estimation.
- Demand for Fuel Economy:
  - Grigolon et al. (2018): Heterogeneous consumer valuation of fuel costs.
  - Leard et al. (2023): Distributional impacts of US fuel economy standards.
- Electric Vehicle Demand:
  - Li et al. (2017) and Springel (2021): Interaction between EV demand and public charging station availability.
  - Bushnell et. al. (2022): Energy prices and electric vehicle adoption

Contribution

- Focuses on EVs, a rapidly growing segment in the US auto market.
- Quantifies how battery capacity and driving range influence EV demand.
- Provides empirical evidence of demographic heterogeneity in preferences for EVs.

Data Sources

- **Wards Automotive** (2014-2024)
  - Model-level data covering all new cars and light trucks sold in the US.
  - Includes sales, prices, and detailed vehicle attributes such as size, weight, horsepower, MPG, fuel type, battery capacity, and driving range.
  - 515 models, 2,851 model-year obs.
- **Consumer Expenditure Survey (CEX)** (2014-2024)
  - Household-level demographics, including income, education, age, and urban/rural status.
  - Records whether a household purchased a new vehicle and the fuel type of the vehicle purchased.

New Vehicle Purchase by Income Quantiles

- Purchase probability ↑ with income.
- High-income HHs are more likely to purchase hybrids and EVs.
- EV purchases shows a steeper income gradient.

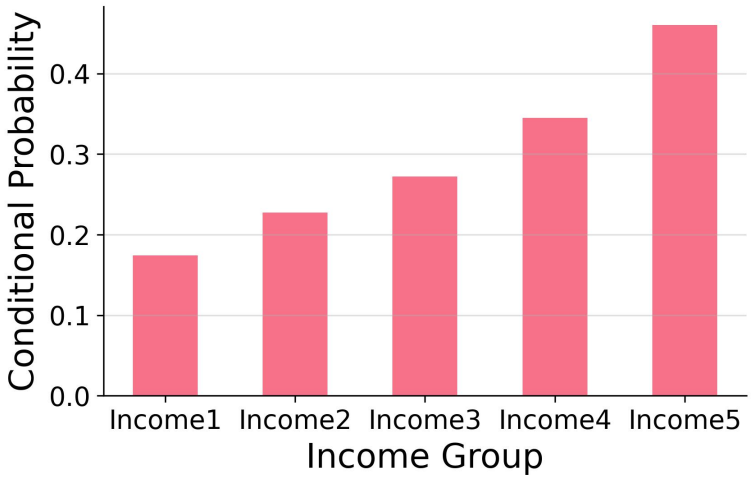


Figure: All New Vehicle Purchase

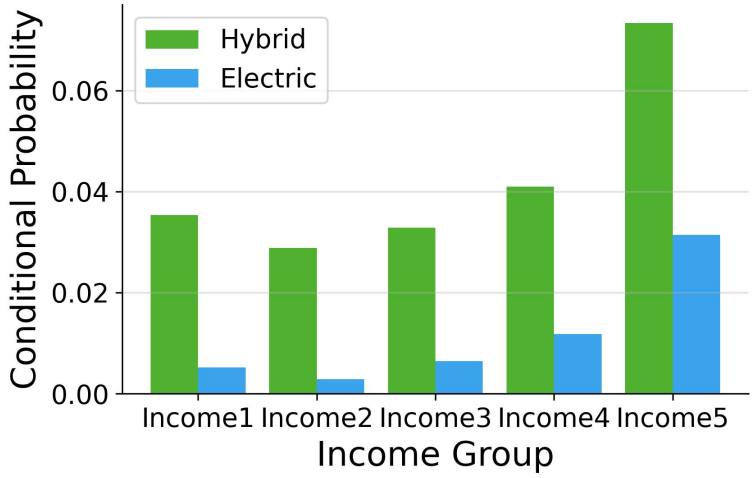


Figure: New Hybrid/EV Purchase

New Vehicle Purchase by Education Level

- Purchase probability is higher for college-educated HHs.
- College-educated HHs are more likely to purchase hybrids and EVs.
- EV purchases shows a steeper education gradient.

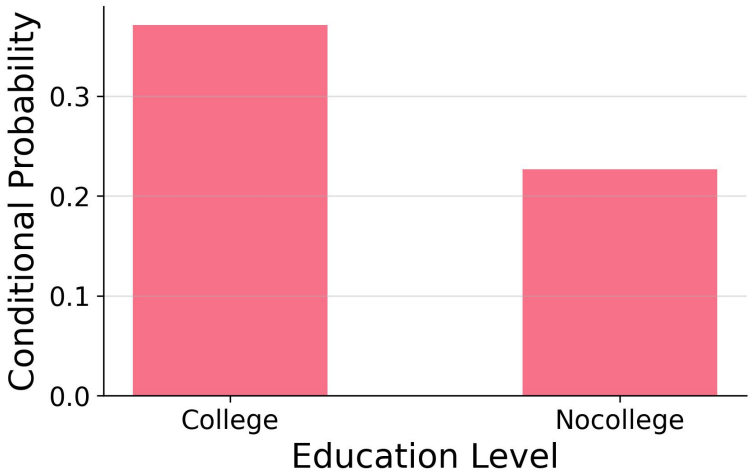


Figure: All New Vehicle Purchase

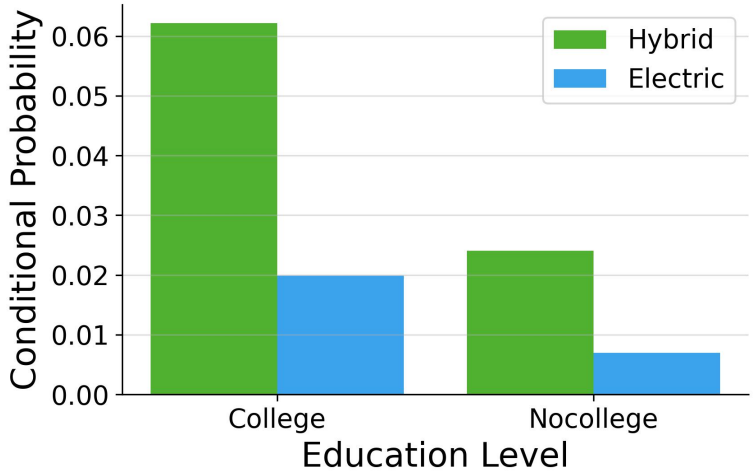


Figure: New Hybrid/EV Purchase

New Vehicle Purchase by Metro Area

- Purchase probability is higher for urban HHs.
- Urban HHs are more likely to purchase hybrids and EVs.
- EV purchases shows a steeper urban-rural status gradient.

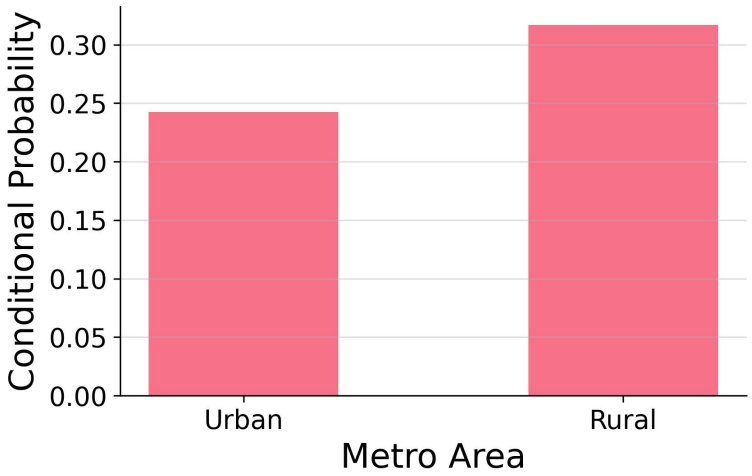


Figure: All New Vehicle Purchase

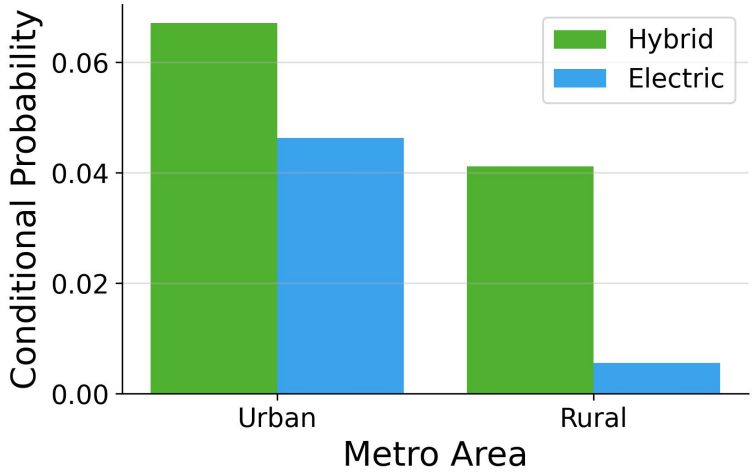


Figure: New Hybrid/EV Purchase

Consumer Utility

- Indirect utility of a consumer i from purchasing a vehicle j in year t is:

$$u_{ijt} = \alpha_i p_{jt} + \beta_i \mathbf{x}_{jt} + \xi_{jt} + \epsilon_{ijt}$$

where

p_j: Vehicle price
$\mathbf{x}_j$: Vehicle attributes (e.g., size, horsepower)
ξ_j: Unobserved vehicle characteristics
ϵ_{ij}: Idiosyncratic consumer-vehicle-specific variation

Heterogeneity in Preferences

- Consumer heterogeneity:

$$\alpha_i = \bar{\alpha} + \sum_h \alpha_h \mathbf{D}_i^h$$

$$\beta_{ik} = \bar{\beta}_k + \sum_h \beta_{kh} \mathbf{D}_i^h + \sigma_k \nu_{ik}$$

where

- h: dimensions of consumer demographics (e.g., income)
- k: dimensions of vehicle characteristics (including a constant)
- ν_{ik}: unobserved taste for automobile characteristics (assumed to be independent draws from N(0,1))
- The distribution of D_i is taken from the CPS.

Estimation Strategy

- Price is instrumented using characteristics of competing products (BLP instruments).
- Included micro-moment conditions.

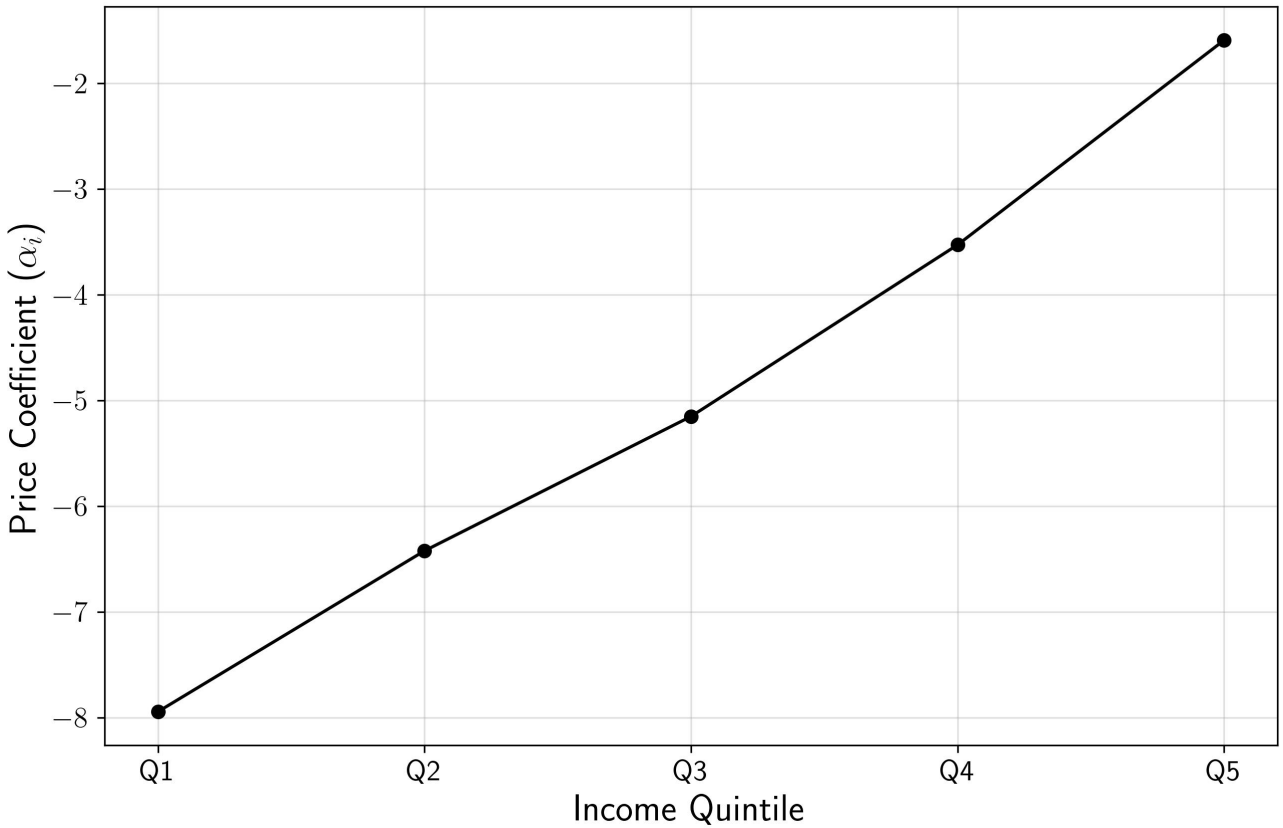
Random Coefficient Estimation

Variable	OLS Logit (1)	IV Logit (2)	Random Coefficients (3)
	A. Price Coefficients (α's)		
α_1	-2.165***	-2.965***	-7.944***
α_2			-6.422**
α_3			-5.150**
α_4			-3.528**
α_5			-1.592**
	B. Base Coefficients (β's)		
Space	0.840***	0.971***	1.072***
Horsepower/Weight	0.212*	0.340**	0.413***
MPG	0.979***	0.861***	1.018***
AWD	0.356**	0.372**	0.512***
SUV	0.304*	0.209	0.316**
Truck	-0.174	-0.530	-0.544
Van	-0.629*	-0.813*	-0.832***
EV	-10.280***	-10.040***	-14.566***
Hybrid	-2.391***	-2.000***	-3.149***
Battery Capacity	0.442**	0.425***	0.299***
Driving Range	1.021*	1.142**	1.228***

Logit Estimation: Interpretations

- **Price:**
  - Negative and significant in both OLS (-2.16) and IV (-2.96).
  - IV estimate is stronger due to correction for endogeneity.
  - Low-income HHs are more price-sensitive.
- **AWD, Space, Horsepower, and MPG:**
  - Positive and significant: HHs prefer AWD, larger vehicles, higher performance, and better fuel economy.
- **EV and Hybrid:**
  - Negative and significant: HHs' disutility relative to gasoline vehicles.
- **Battery Capacity and Driving Range:**
  - Positive and significant: HHs value these features, which partly offset EV disutility.

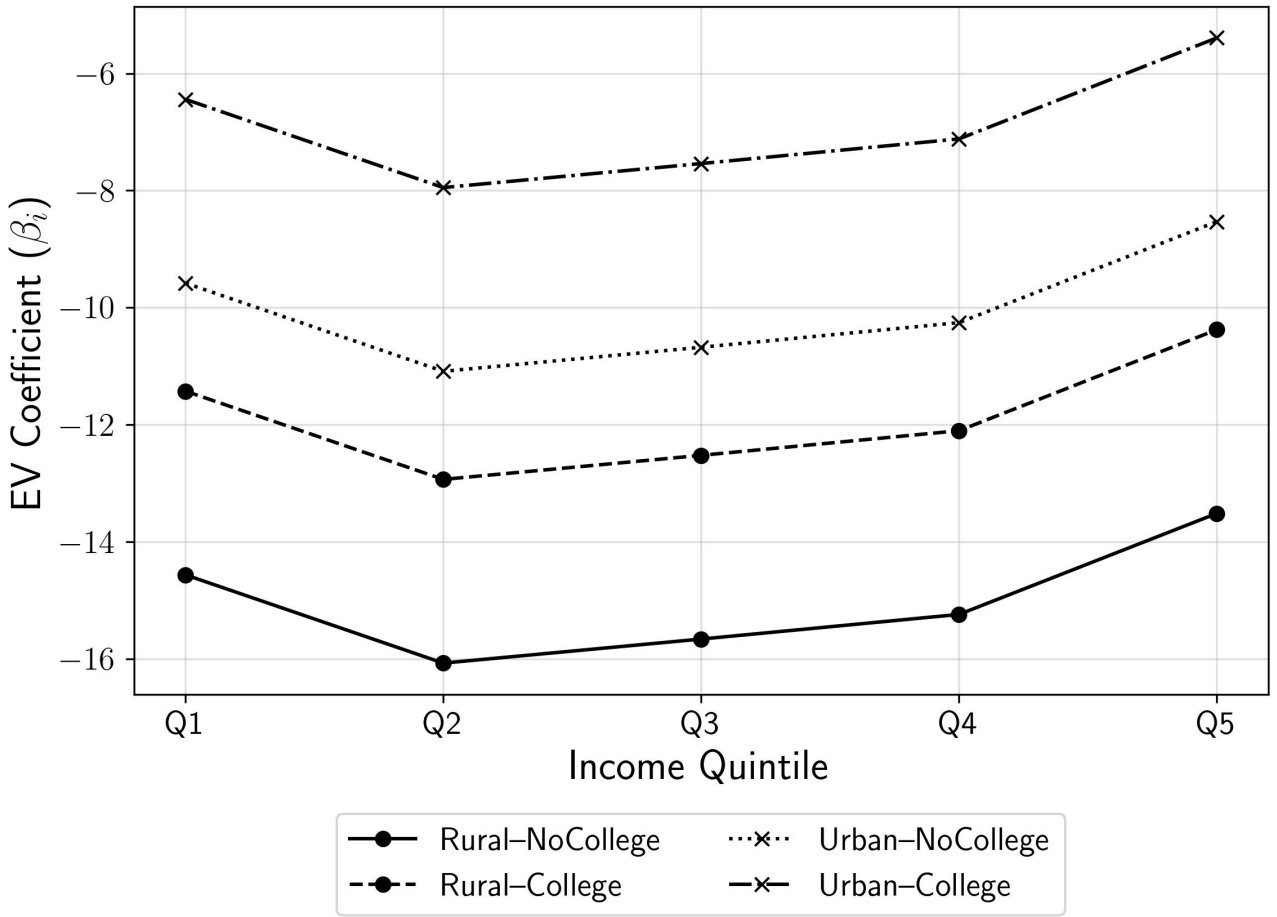
Price Coefficients



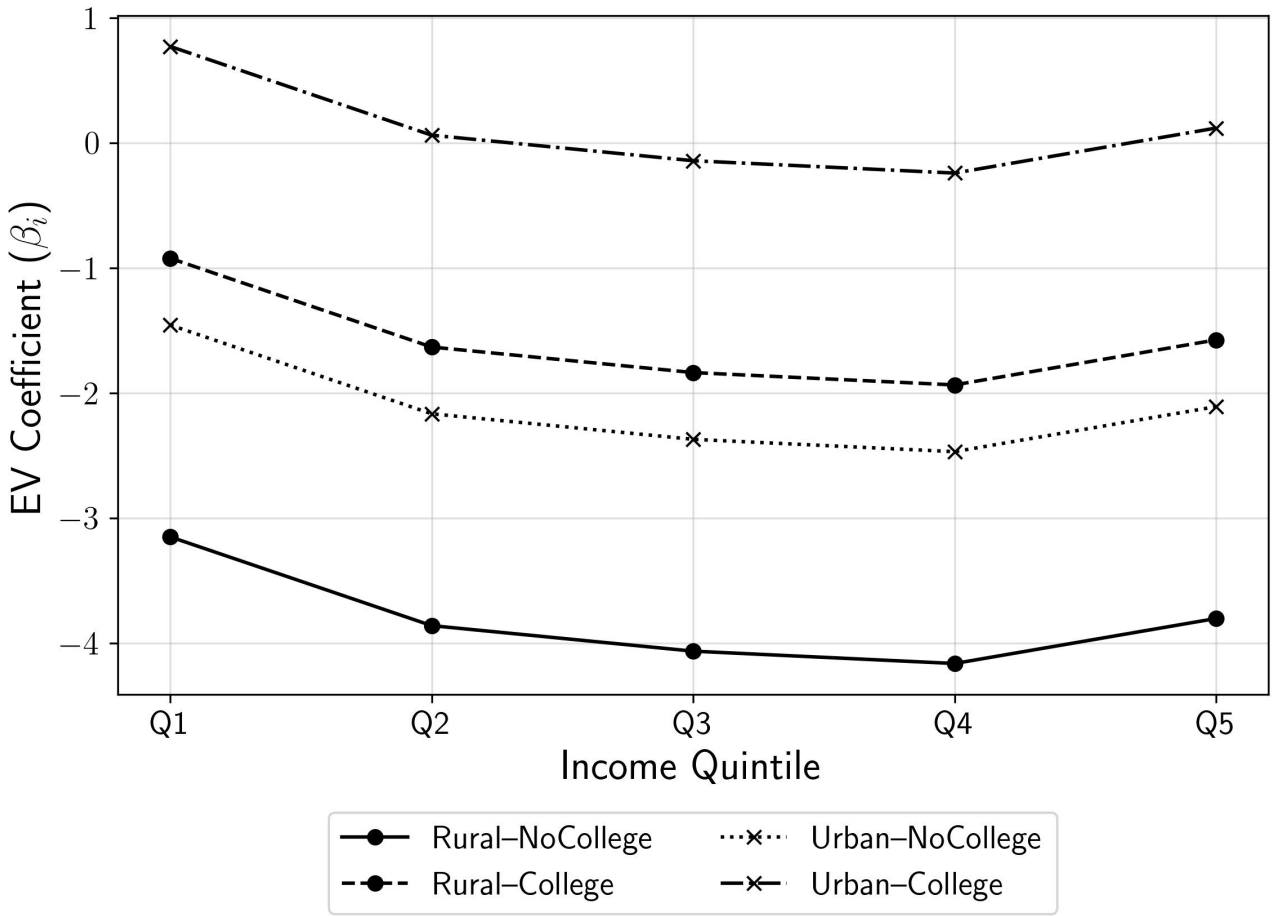
Random Coefficient Estimation

Variable	EV (1)	HEV (2)
	A. Baseline (Q1 Income, Rural, Non-College)	
β	-14.566***	-3.149**
	B. Income Adjustments	
Q2 Income Adjustments	-1.508***	-0.709***
Q3 Income Adjustments	-1.097***	-0.914***
Q4 Income Adjustments	-0.676	-1.012***
Q5 Income Adjustments	1.050***	-0.652*
	C. Urban/College Adjustments	
Urban Adjustments	4.985***	1.693***
College Adjustments	3.139***	2.226***

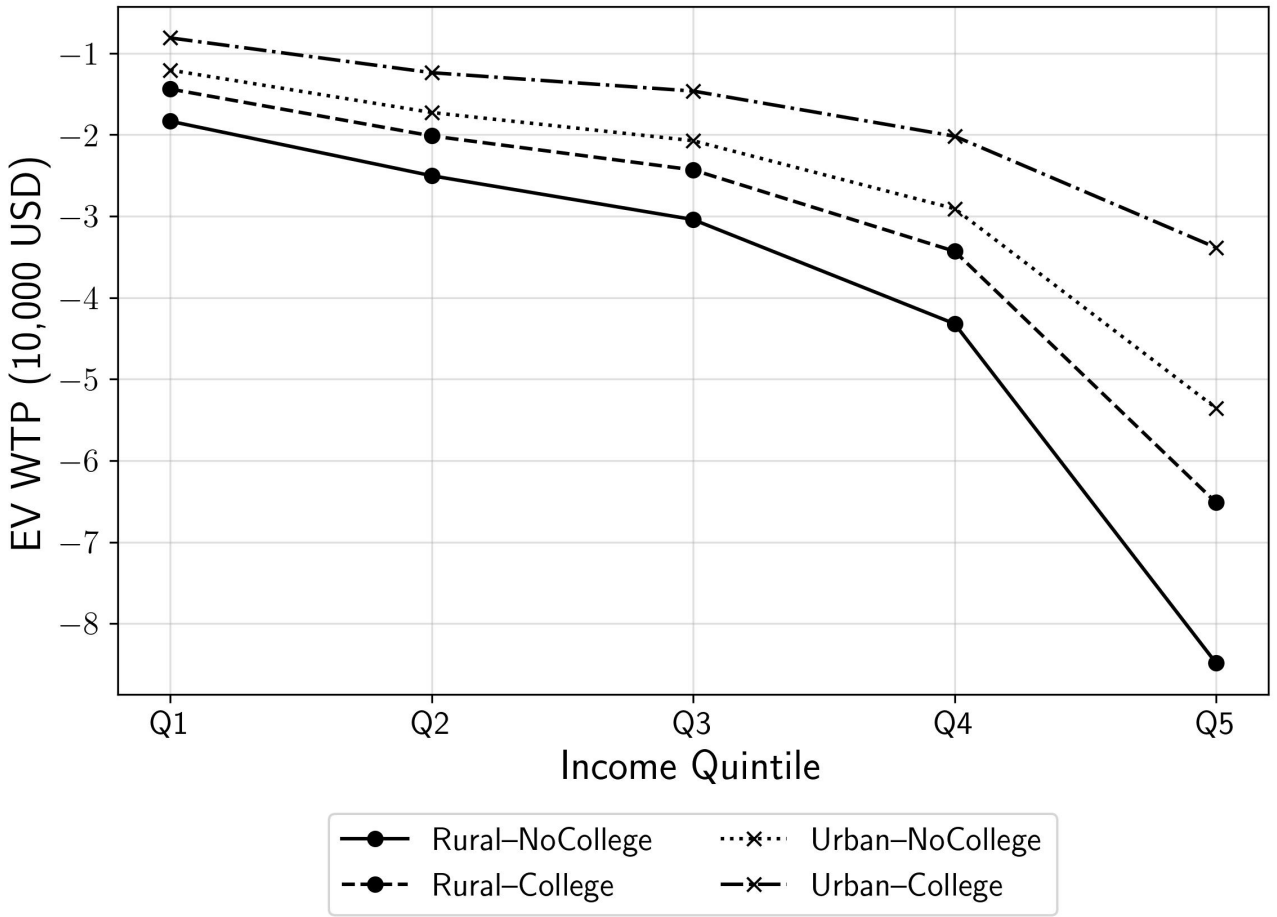
EV Coefficients



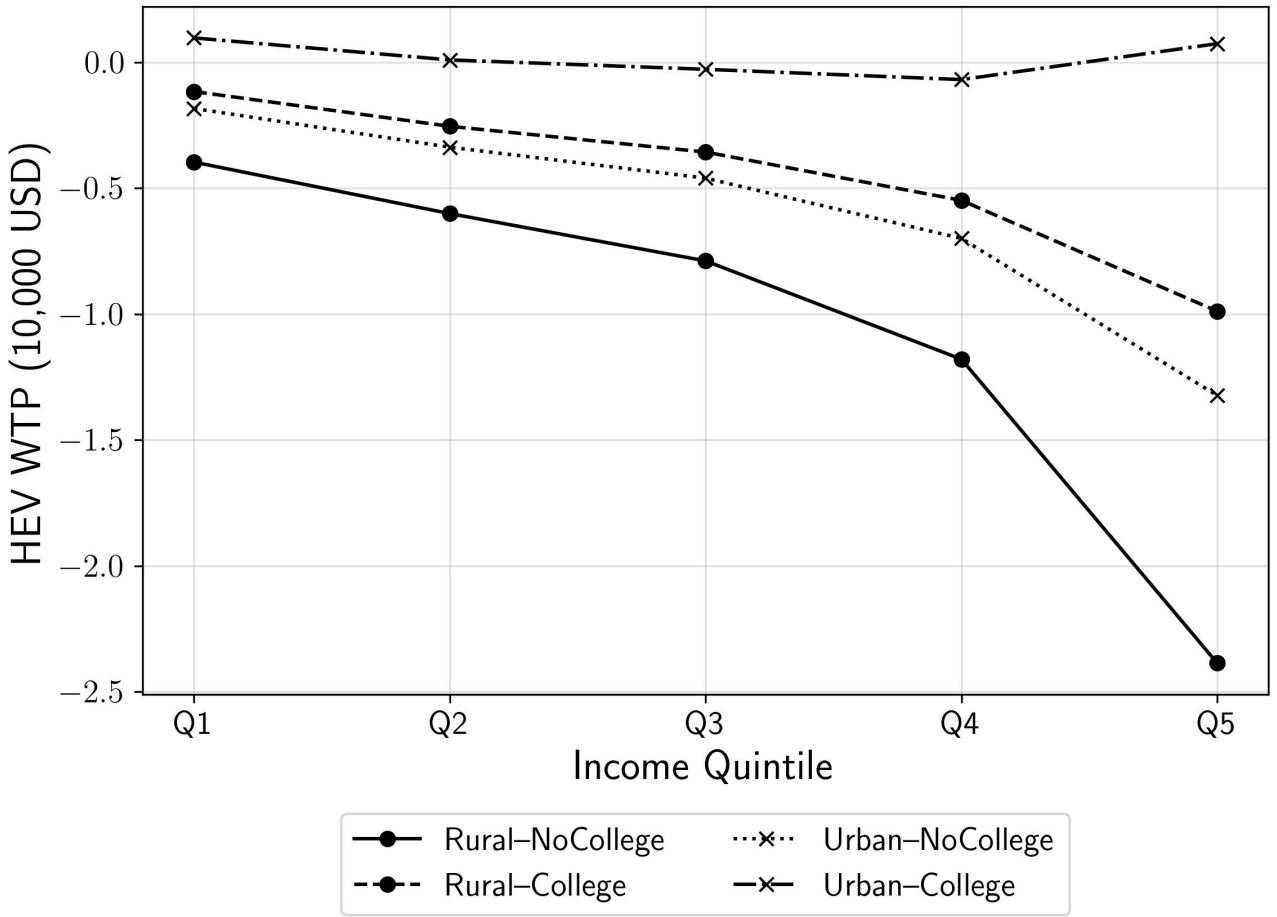
HEV Coefficients



EV WTP


EV WTP (10,000 USD)
−1
−2
−3
−4
−5
−6
−7
−8
Q1
Q2
Q3
Q4
Q5
Income Quintile
Rural–NoCollege
Rural–College
Urban–NoCollege
Urban–College

HEV WTP


HEV WTP (10,000 USD)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
Q1
Q2
Q3
Q4
Q5
Income Quintile
Rural–NoCollege
Urban–NoCollege
Rural–College
Urban–College

Conclusion

- Consumers exhibit significant disutility for EVs and hybrids relative to gasoline vehicles.
- Technical features like battery capacity and driving range are positively valued and help offset baseline EV disutility.
- EV preferences are highly heterogeneous across demographic groups:
  - High-income consumers are less price-sensitive.
  - High-income groups show strong EV preference.
  - Urban and college-educated groups show strong EV preference.
- These results highlight the need for targeted EV policies informed by heterogeneous demand.

Estimated Micro Moments

Estimated Micro Moments:

Observed	Estimated	Difference	Moment
+3.014E-02	+3.018E-02	-4.393E-05	E[income2 | electric]
+7.941E-02	+7.947E-02	-5.922E-05	E[income3 | electric]
+1.868E-01	+1.869E-01	-3.714E-05	E[income4 | electric]
+6.625E-01	+6.622E-01	+2.446E-04	E[income5 | electric]
+7.073E-01	+7.075E-01	-2.139E-04	E[urban | electric]
+8.127E-01	+8.126E-01	+6.094E-05	E[college | electric]
+9.373E-02	+9.369E-02	+3.552E-05	E[income2 | hybrid]
+1.277E-01	+1.277E-01	+5.239E-05	E[income3 | hybrid]
+2.035E-01	+2.034E-01	+8.491E-05	E[income4 | hybrid]
+4.865E-01	+4.867E-01	-1.826E-04	E[income5 | hybrid]
+3.223E-01	+3.223E-01	-6.432E-07	E[urban | hybrid]
+7.969E-01	+7.969E-01	-2.802E-05	E[college | hybrid]
+2.276E-01	+2.276E-01	-6.294E-06	E[purchase | income2]
+2.720E-01	+2.721E-01	-7.095E-06	E[purchase | income3]
+3.453E-01	+3.453E-01	-3.820E-05	E[purchase | income4]
+4.604E-01	+4.604E-01	+2.541E-05	E[purchase | income5]